Indo-Quantum Summit Gujarat

Peaked Quantum Circuits and Classical Inferability of Dominant Bitstrings

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Brief Motivation

Quantum advantage via sampling faces a **verification challenge**, i.e., it is non-trivial to classically determine whether the output of a circuit is correct. However, one such proposed property of the output to weakly verify the evidence of correct behavior is the presence of a **dominant bitstring.**

If a circuit's distribution is sharply concentrated, the peak can serve as a concise classical certificate. These circuits are known as **peaked circuits**. Recent results show that generic or random quantum circuits almost never produce such peaks, implying that any dominant bitstring must arise from added **circuit structure**. We then study the **different classes** of peaked quantum circuits to best understand how to utilize them advantageously.

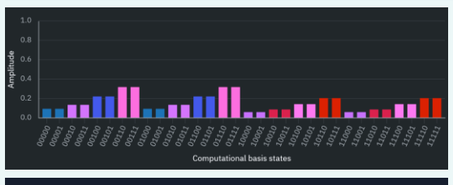


Fig 1. Random Quantum Circuit

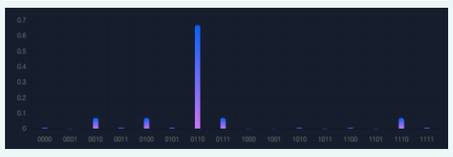


Fig 2. Peaked Quantum Circuit

Objective

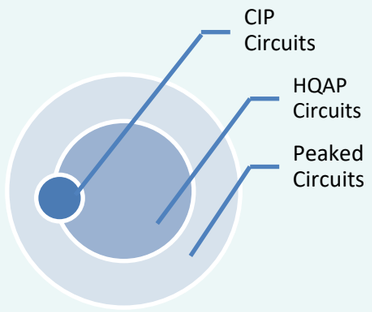


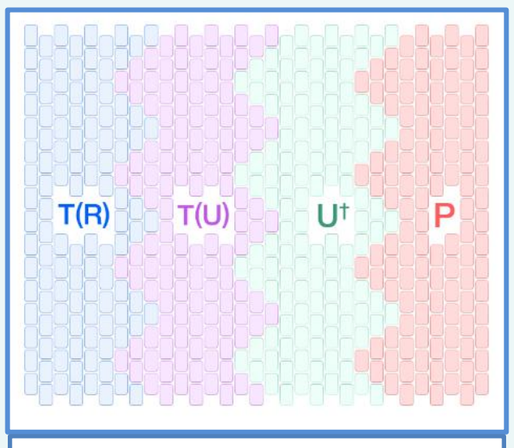
Fig 3. Classes of peaked circuits represented through a Venn diagram. Note that this diagram is only to explore the existence of CIP circuits and may not represent the hierarchy accurately until further work is performed in the area.

The focus of this study is to identify and categorize **embedded substructure** in peaked quantum circuits. Previous works show that many such circuits are classically intractable to simulate on consumer-grade hardware, claiming potential cryptographic advantages linked to the "**quantum supremacy**" goal. We study defined HQAP circuits [1] and **CIP circuits** to learn more about how added structure makes inference classically tractable. The objective is to fill in gaps in existing literature on this topic.

Methodology

Circuit Definition → Structural Analysis → Inferability Assessment

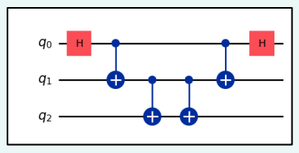
Q. How can PQCs be constructed from a generic random circuit?



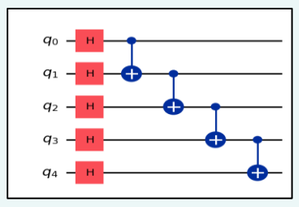
Visual schematic of HQAP building. Refer to [1] for circuit manipulations used to scramble the circuit structure.

- Train a shallow peaked circuit R ▷ P using the method in Ref. [3].
- Insert an identity of the form U ▷ U † between R and P .
- Apply sweeping to variational circuit parameters.
- Apply masking to modify the connectivity and gate structure.
- Apply swap transformations throughout the circuit.

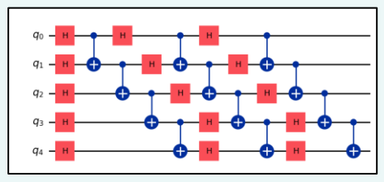
Static Circuit Structure Analysis & Determination of Classical Inferability



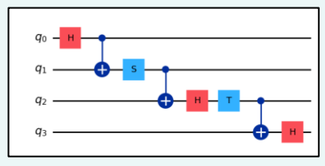
Mirror (symmetry)
Mirror or time-reversal symmetry circuits' second half approximately inverts or reflects the first. Such symmetry strongly constrains the output state and can force constructive interference on a specific computational basis string.



Near-neighbor entanglement
Circuits with such structure that show mainly local entanglement restrict the spread of quantum correlation. This can leave residual classical structure, allowing the dominant output to be classically inferred by exploiting locality and causal constraints in the system.



Repeating patterns
Repeated gate motifs or periodic subcircuits break the statistical independence expected of random quantum circuits. This can introduce deterministic correlations that bias the output toward specific bitstrings, and when visible in the circuit description, may enable classical inference of the dominant output.



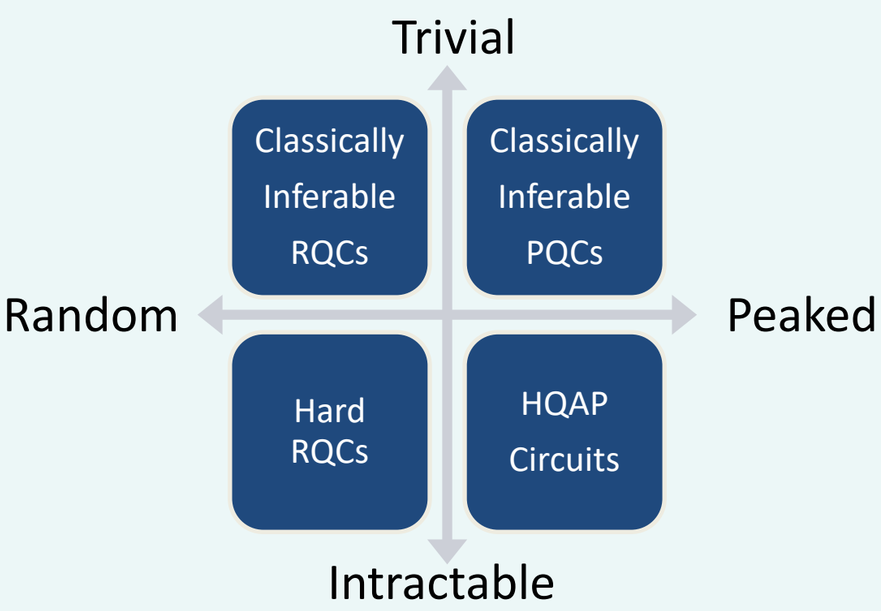
Embedded classical behavior or symmetry results in inferable peaks.

Note: These circuit examples only serve to demonstrate the structural patterns observed in many common peaked circuits.

Does the state output by a polynomial-size RQC have any "**exploitable structure**" at all to distinguish it from a Haar-random state? Ref. [2]

Numerical results from **Aaronson** (Ref. [2]) provide strong **evidence** to the existence of such structure. Further study reveals that it is possible to **achieve non-trivial peakedness** with a small number of additional gates, strongly pointing towards the possibility of revealing certain dominant bitstrings using classical inference methods. We classify such circuits as **CIP** (classically inferable peaked) circuits and are different from **HQAP** (heuristic quantum advantage with peaked) circuits.

Results



Note: This diagram shows the various classes of quantum circuits classified based on randomness of the probability distribution and the ease of sampling (or XEB) for validation of behavior.

1. Peakedness of a quantum state is **insufficient** for verification.
2. Existing literature addresses the existence of a peak, but its **inferability** through classical means has not been studied yet. This work proposes families to categorize circuits based on their structure.
3. PQCs can be **classified** into HQAP circuits and CIP circuits based on structure and ease of classical validation.

Diagnostic directions that we analyze show how deviations from generic quantum behavior (such as reduced output entropy, violations of anticoncentration, simple correlation patterns, limited entanglement growth, or departures from approximate randomness) constrain the output distribution. This suggests that many peaked circuits might have classical substructure, requiring the need of a more refined criterion when verifying and benchmarking results in NISQ era quantum experiments.

The present work is concerned with a distinct question: whether the structure responsible for inducing peakedness also enables efficient classical inference of the dominant output string from the circuit description alone. These two notions of structure need not coincide. It is possible for circuits to contain sufficient structure to allow peaks to form, while still concealing the location of the peak from classical static analysis. Clarifying this distinction separates structural explanations of peak formation from algorithmic criteria for classical inferrability.

Conclusion

This work studies the **two classes** of peaked quantum circuits to clarify previous related papers further into verifiable quantum advantage. Papers like **Aaronson et al. (2024)** and **Gharibyan et al. (2025)** establish the foundations of peaked behavior arising from random circuits containing hidden structure and construction of **HQAP** circuits. This work is important as determining whether a circuit is peaked given only the input and output bitstrings is a **QCMA** class problem and can be used to claim **cryptographic** quantum advantage.

Future work aims to **formalize distinguishing criteria** for inferable and non-inferable peaked circuits, to better understand which structural features expose or **conceal dominant outputs**, and to assess the implications of this distinction for verification and benchmarking protocols in the **NISQ era**.

References

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1. Hrant Gharibyan et al. *Heuristic Quantum Advantage with Peaked Circuits.* 2025.

2. Scott Aaronson, Yiqing Liu, Alexei Nesterov, and John Wright. *Peaked output distributions in random quantum circuits*. 2024.

3. Oleg Udalov. *A Method for Constructing Quasi-Random Peaked Quantum Circuits*. 2025.